

Api Structure V2

API Design Structure

I would organize the APIs into:

```
/api/v1
```

```
|— auth  
|— users  
|— students  
|— employees  
|— courses  
|— attendance  
|— work-reports  
|— payroll  
|— fees  
|— crm  
|— whatsapp  
|— recruitment  
|— blogs  
|— dashboard  
|— notifications  
|— uploads
```

1. AUTH MODULE

Authentication

Login

```
POST /api/v1/auth/login
```

Body

```
{  
  "email": "john@gmail.com",
```

```
"password": "123456"
}
```

Response

```
{
  "accessToken": "",
  "refreshToken": "",
  "user": {}
}
```

Refresh Token

POST /api/v1/auth/refresh

Logout

POST /api/v1/auth/logout

Change Password

PUT /api/v1/auth/change-password

Forgot Password

POST /api/v1/auth/forgot-password

2. USERS MODULE

Common user management.

Get Current User

```
GET /api/v1/users/me
```

Update Profile

```
PUT /api/v1/users/me
```

Upload Profile Image

```
POST /api/v1/users/profile-image
```

3. STUDENT MANAGEMENT APIs

(PRD Section 4.1)

Create Student

```
POST /api/v1/students
```

List Students

```
GET /api/v1/students
```

Filters

```
?page=1  
&limit=20  
&courseId=
```

```
&status=  
&search=
```

Get Student

```
GET /api/v1/students/:studentId
```

Update Student

```
PUT /api/v1/students/:studentId
```

Delete Student

```
DELETE /api/v1/students/:studentId
```

Assign Course

```
POST /api/v1/students/:studentId/courses
```

Student Progress

```
GET /api/v1/students/:studentId/progress
```

Student Reports

GET /api/v1/students/reports

Student Documents

POST /api/v1/students/:studentId/documents

GET /api/v1/students/:studentId/documents

DELETE /api/v1/students/:studentId/documents/:documentId

4. EMPLOYEE MANAGEMENT APIs

(PRD Section 4.2)

Create Employee

POST /api/v1/employees

Get Employees

GET /api/v1/employees

Get Employee

GET /api/v1/employees/:employeeId

Update Employee

PUT /api/v1/employees/:employeeId

Employee Documents

POST /api/v1/employees/:employeeId/documents

Assign Course

POST /api/v1/employees/:employeeId/courses

Employee Report

GET /api/v1/employees/reports

5. COURSE MANAGEMENT APIs

(PRD Section 4.3)

Create Course

POST /api/v1/courses

Get Courses

GET /api/v1/courses

Get Course

GET /api/v1/courses/:courseId

Update Course

PUT /api/v1/courses/:courseId

Publish Course

PATCH /api/v1/courses/:courseId/publish

Archive Course

PATCH /api/v1/courses/:courseId/archive

Course Students

GET /api/v1/courses/:courseId/students

Course Employees

```
GET /api/v1/courses/:courseId/employees
```

6. ATTENDANCE APIs

(PRD Section 4.4)

Mark Attendance

```
POST /api/v1/attendance
```

Body

```
{
  "userId": "",
  "date": "",
  "status": "present"
}
```

Bulk Attendance

```
POST /api/v1/attendance/bulk
```

Attendance History

```
GET /api/v1/attendance
```

Filters

```
?userId=
&from=
```


&to=

Attendance Report

GET /api/v1/attendance/reports

7. WORK REPORT APIs

(PRD Section 4.5)

Submit Report

POST /api/v1/work-reports

Get Reports

GET /api/v1/work-reports

Get Single Report

GET /api/v1/work-reports/:reportId

Approve Report

PATCH /api/v1/work-reports/:reportId/approve

Reject Report

```
PATCH /api/v1/work-reports/:reportId/reject
```

8. PAYROLL APIs

(PRD Section 4.6)

Generate Payroll

```
POST /api/v1/payroll/process
```

Payroll History

```
GET /api/v1/payroll
```

Payroll Details

```
GET /api/v1/payroll/:payrollId
```

Download Payslip

```
GET /api/v1/payroll/:payrollId/payslip
```

9. FEES APIs

Record Payment

POST /api/v1/fees/payments

Student Payments

GET /api/v1/fees/students/:studentId

Receipt

GET /api/v1/fees/payments/:paymentId/receipt

Outstanding Fees

GET /api/v1/fees/outstanding

10. SALES CRM APIs

(PRD Section 4.7)

Create Lead

POST /api/v1/crm/leads

Get Leads

```
GET /api/v1/crm/leads
```

Lead Details

```
GET /api/v1/crm/leads/:leadId
```

Update Stage

```
PATCH /api/v1/crm/leads/:leadId/stage
```

Add Interaction

```
POST /api/v1/crm/leads/:leadId/interactions
```

Follow Up

```
POST /api/v1/crm/leads/:leadId/followups
```

Convert To Student

```
POST /api/v1/crm/leads/:leadId/convert
```

11. RECRUITMENT APIs

(PRD Section 4.9)

Create Candidate

POST /api/v1/recruitment/candidates

Upload Resume

POST /api/v1/recruitment/candidates/:candidateId/resume

Schedule Interview

POST /api/v1/recruitment/interviews

Candidate Pipeline

PATCH /api/v1/recruitment/candidates/:candidateId/stage

Candidate Board

GET /api/v1/recruitment/board

12. BLOG DASHBOARD APIs

(PRD Section 4.10)

Blog Stats

```
GET /api/v1/blogs/stats
```

Blog List

```
GET /api/v1/blogs
```

Top Blog

```
GET /api/v1/blogs/top-performing
```

13. DASHBOARD APIs

This is what powers the UI home page.

Admin Dashboard

```
GET /api/v1/dashboard/admin
```

Returns:

```
{
  "students":120,
  "employees":15,
  "activeCourses":8,
  "monthlyRevenue":450000,
  "attendancePercentage":92
}
```

Sales Dashboard

```
GET /api/v1/dashboard/sales
```

HR Dashboard

```
GET /api/v1/dashboard/hr
```

Marketing Dashboard

```
GET /api/v1/dashboard/marketing
```

Employee Dashboard

```
GET /api/v1/dashboard/employee
```

14. FILE UPLOAD APIs

Since you're using Cloudinary:

```
POST /api/v1/uploads
```

For:

- Student documents
- Employee documents
- Resume uploads
- Profile images
- Payroll slips

- Certificates

What I Would Build First

Do **not** start with all APIs.

Build in this order:

Phase 1 (MVP)

Auth
Users
Students
Employees
Courses
Attendance
Dashboard

Phase 2

Work Reports
Fees
Payroll
CRM

Phase 3

Recruitment
Blog Dashboard
WhatsApp Automation
RAG Chatbot
n8n Integrations